

Development Session Log

Powerweave Smart Task Escalation Engine

Date: 26 May 2026

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Session: Full Build & Deploy

Status: COMPLETE

66
Tests Passing

5
Test Suites

14
Files Committed

6
DB Tables Created

3
API Endpoints Live

1
PR Merged

1. SESSION OBJECTIVES

Bootstrap the Powerweave Smart Task Escalation Engine from source files to a fully deployed, tested, and documented REST API backed by a cloud PostgreSQL database.

- Resolve missing build artefacts and TypeScript compilation errors
- Set up Prisma schema, generate client, and run migrations against Neon (PostgreSQL)
- Seed the database with representative test data
- Wire up Express routes with RBAC middleware and test all endpoints
- Achieve full CI-green test suite (unit tests across all modules)
- Establish Git workflow: init → commit → PR → merge

2. ISSUES RESOLVED

#	Issue	Root Cause	Resolution	Status
1	Cannot find module 'dist/index.js'	Missing tsconfig.json , src/index.ts entry point, and Prisma schema	Created tsconfig.json , src/index.ts , prisma/schema.prisma , ran npm run build	Fixed
2	TypeScript: Property 'user' does not exist on Request	No Express type augmentation for req.user	Created src/types/express.d.ts extending Express Request	Fixed

3	Prisma: <code>EscalationStatus</code> , <code>EscalationTier</code> not exported	Prisma client not generated (schema missing)	Created <code>prisma/schema.prisma</code> with all models and enums, ran <code>prisma generate</code>	Fixed
4	Test TS error: <code>'()' => string'</code> not assignable to <code>Date</code> mock	Jest Date mock cast used <code>as unknown as string</code> instead of <code>Date</code>	Changed cast to <code>as unknown as Date</code> in both affected test files	Fixed
5	Audit log test: <code>oldValue: JsonNull {}</code> vs expected <code>null</code>	Code used <code>Prisma.JsonNull</code> sentinel (an object), tests expected JS <code>null</code>	Changed cast to <code>(oldValue ?? null) as unknown as Prisma.InputJsonValue</code>	Fixed
6	POST <code>/escalations</code> returning <code>VALIDATION_ERROR</code> for taskld	Seed data used non-UUID IDs (e.g. <code>task-00000-0000-...</code>)	Rewrote seed with proper UUID format (<code>00000000-0000-0000-0000-000000000021</code>)	Fixed
7	API endpoints returning 401 for all requests	No JWT auth middleware implemented to populate <code>req.user</code>	Added dev auth middleware reading from <code>X-User-*</code> headers	Fixed
8	Prisma <code>EPERM</code> on DLL rename during generate	Node process held file lock on <code>query_engine-windows.dll.node</code>	Killed stale Node processes before running <code>prisma generate</code>	Fixed
9	GitHub PR creation returning 403	Fine-grained PAT lacked explicit <code>pull_requests: write</code> permission	Switched to Classic PAT with <code>repo</code> scope	Fixed
10	PR branches had no shared history (422 from GitHub API)	<code>main</code> was created as an orphan branch	Recreated <code>main</code> from <code>master</code> HEAD so both share history	Fixed

3. ARTEFACTS CREATED

File / Artefact	Type	Purpose
<code>tsconfig.json</code>	Config	TypeScript compiler settings (rootDir: src, outDir: dist, strict)
<code>src/index.ts</code>	Source	Express app entry point — routes, middleware, error handler
<code>src/types/express.d.ts</code>	Type def	Augments Express Request with <code>req.user</code> shape
<code>prisma/schema.prisma</code>	Schema	Prisma schema for all 6 models: User, Team, Task, Escalation, EscalationAuditLog, Notification
<code>prisma/migrations/20260526170100_init/</code>	Migration	Initial DB migration — creates all tables in Neon PostgreSQL
<code>prisma/seed.ts</code>	Script	Idempotent seed: 2 teams, 5 users, 3 tasks, 3 escalations, 3 audit logs
<code>.gitignore</code>	Config	Excludes node_modules, dist, .env, coverage
<code>.env</code>	Secret	DATABASE_URL pointing to Neon; gitignored

<code>.env.example</code>	Template	Environment variable template for onboarding
<code>tests/unit/escalation.service.test.ts</code>	Test	22 unit tests for EscalationService (createEscalation, updateStatus, getHistory)
<code>api-schema/escalation.json</code>	Schema	OpenAPI 3.0 JSON schema converted from existing YAML
<code>scripts/generate-session-log.ts</code>	Script	This PDF generator (Playwright-based)

4. DATABASE SETUP

Provider

Neon (serverless PostgreSQL) — free tier, region: `us-east-1`

Tables Created

Table	Rows (seed)	Key Columns
<code>teams</code>	2	id, name, managerId
<code>users</code>	5	id, email, name, role, teamId
<code>tasks</code>	3	id, title, teamId
<code>escalations</code>	3	id, taskId, tier, status, escalatedBy, assignedManager, resolvedAt, resolutionNote
<code>escalation_audit_logs</code>	3	id, escalationId, action, performedBy, oldValue, newValue, timestamp
<code>notifications</code>	0	id, escalationId, recipientId, channel, status, sentAt

Migration Command

```
npx prisma migrate dev --name init
```

Seed Command

```
npx ts-node prisma/seed.ts
```

5. API ENDPOINTS — LIVE TEST RESULTS

Method	Path	Role Required	Result	HTTP
POST	<code>/api/v1/escalations</code>	ADMIN, MANAGER	Created escalation <code>2f66dacb...</code> , auto-assigned to Team Beta manager, notification dispatched	201
GET	<code>/api/v1/escalations/:id/history</code>	All roles (team-scoped)	Returned 2 audit log entries (CREATED + STATUS_CHANGED) with pagination metadata	200
PATCH	<code>/api/v1/escalations/:id/status</code>	ADMIN, MANAGER	Transitioned escalation from <code>OPEN</code> → <code>IN_PROGRESS</code> with audit log written	200
POST	<code>/api/v1/escalations</code> (no auth)	—	RBAC guard correctly rejected unauthenticated request	401

6. TEST SUITE RESULTS

Test File	Tests	Coverage Area	Result
<code>escalation.service.test.ts</code>	22	createEscalation, updateStatus, getHistory — all business logic paths	PASS
<code>escalation.validation.test.ts</code>	12	Zod schemas: CreateEscalationSchema, UpdateStatusSchema — boundary values	PASS
<code>role.permission.test.ts</code>	10	RBAC permission map, requirePermission middleware	PASS
<code>notification.trigger.test.ts</code>	10	NotificationService.dispatch — EMAIL, IN_APP, overrides, SES failure	PASS
<code>audit.logger.test.ts</code>	12	AuditLogger.log, getLogsForEscalation — non-fatal error handling	PASS
Total: 66 tests		5 suites — 0 failures — ~5.5s	

7. GIT HISTORY

Commit	Message	Branch
719fa68	Initial commit: Powerweave escalation engine (29 files)	master
3b4de1d	chore: add .env.example with required environment variables	master
0fd614e	feat: Add Powerweave escalation engine (<i>PR #1 merge commit</i>)	main
6475dbb	chore: add initial Prisma migration	main
9108423	chore: add database seed with teams, users, tasks and escalations	main
d56dcc5	feat: add dev auth middleware and fix seed UUIDs for API testing	main

8. GITHUB REPOSITORY

Item	Detail
Repository	https://github.com/Manas-Naik/powerweave-escalation
Default branch	<code>main</code>
PR #1	feat: Add Powerweave escalation engine — Merged
CI	Not configured (recommended: add GitHub Actions for <code>npm test</code> on push)

9. NEXT STEPS / RECOMMENDATIONS

- **Replace dev auth middleware** with proper JWT verification (`jsonwebtoken` already in dependencies)
- **Upgrade Node.js to v22+** to eliminate AWS SDK version warning
- **Add GitHub Actions CI** — run `npm test` on every push to `main`
- **Configure AWS SES** — set real `AWS_REGION` and `SES_FROM_EMAIL` in production env
- **Revoke exposed GitHub tokens** — both PATs shared during this session should be deleted
- **Add integration tests** — Supertest against a test database for end-to-end route coverage
- **Enable Prisma migrations in CI** — run `prisma migrate deploy` on staging/production deploy

